



Making Research Software Citable & Discoverable

UC Carpentries Workshop Series · Fall 2026

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Welcome!

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We'll begin shortly. Grab the collaborative notes and settle in.

Etherpad: pad.carpentries.org/2026-fall-uc-sharing-research-software

Lesson site: ucospo.net/research-software-citable-discoverable

Code of Conduct

We agree to

- Use **welcoming and inclusive** language
- Be **respectful** of different viewpoints and experiences
- Accept **constructive criticism** gracefully
- Show **courtesy and respect** toward everyone here

If something's wrong

- Tell an **instructor** or the **host** — chat DM or email, privately
- Or report to the **CoC Committee**:
coc@carpentries.org
- All reports are **kept confidential** and taken seriously

Full text & report form: docs.carpentries.org/policies/coc

How we learn together

Our pedagogy

- **Hands-on:** you do each step in your own fork as we go
- **Connect to what you know:** new skills land faster that way
- **Fewer concepts, done well:** not a race

Managing expectations

- **Usable today:** practical skills, not full coverage
- **Group pace:** your questions set the speed
- **Mistakes are expected:** for all of us

The question that starts this lesson

“The journal accepted my paper, but they want a citable version of the code, with a DOI. How do I do that?”

Ten years ago the same question arrived about **datasets**, and libraries built the answer: a persistent identifier, a catalog record, a citation format, a preservation copy. Software mostly hasn't gotten that treatment yet.

You write research software, and want to make your own code citable, reusable, and discoverable.

You support people who do, as a librarian, OSPO member, or research-support specialist.

Supporting others slides translate each step into consultation work.

How common is the software citation gap?

59.5%

of 3.2 million PubMed Central articles mention at least one piece of software

38%

of 2020 articles give enough detail to **uniquely identify** the software they used

Both figures: Schindler et al. (2022), PeerJ Computer Science, doi:10.7717/peerj-cs.835

A GitHub URL isn't a citation

✗ Wrong

“We used the analysis script from github.com/UC-OSPO-Network/software-demo”

- Links rot
- No version: which code produced the result?
- No author, so no credit

✓ Right

Dennis, T., Phan, L., Otsuji, R., & Padilla, K. (2026). *Biodiversity Analysis Toolkit* (v0.1.0). Zenodo.
<https://doi.org/10.5281/zenodo.123456>

- The DOI outlives the repo
- The version is explicit
- Authors get credit and citation counts

You'll create both pieces today: a `CITATION.cff` file and a DOI.

Where your fork is headed

Before (`main`)

-  No LICENSE
-  No citation information
-  No release or DOI
-  Minimal README

After (`05-add-metadata`)

-  LICENSE (BSD-3-Clause) · *Reusable*
-  CITATION.cff · *Findable*
-  Release + DOI on Zenodo · *Findable, Accessible*
-  Full README, CONTRIBUTING, Code of Conduct · *Reusable*

Today: 2 License → 3 CITATION.cff → 4 Release + DOI → 5 Metadata → wrap-up

The labels come from FAIR4RS: Findable, Accessible, Interoperable, Reusable. Check your work any time against the matching reference branch (`02-add-license` ... `05-add-metadata`) in the upstream repo.

Fork the demo repository

Every hands-on step happens in **your own fork**, not the shared demo repo.

1. Open `github.com/UC-OSP0-Network/software-demo`
2. Click **Fork** → **Create fork** (defaults are fine — `main` branch only)
3. Confirm the top-left reads `YOUR-USERNAME/software-demo` , not `UC-OSP0-Network/software-demo`

Reference branches are there to check your work, never to build on. Full setup details: ucospo.net/research-software-citable-discoverable/learners/setup

The starting point

```
software-demo/  
├── README.md  
├── requirements.txt  
└── src/  
    └── analysis.py
```

Deliberately bare. By the end it will have a license, citation metadata, a DOI, and a real README.

Supporting others: a four-question triage

Public is not the same as shared. When a researcher brings you their repo, do a fast triage, not a code review.

1. **Rights** — is there a license? Can this legally be reused at all?
2. **Credit** — can a stranger tell who made it and how to cite it?
3. **Runnable** — is there any record of what it takes to run?
4. **Findable** — would anyone outside the lab ever discover it?

The goal is the **single highest-value next step**, not fixing everything.

Try it now (2 min): answer all four for your fork of [software-demo](#) .

Key points

- A GitHub URL is not a citation. A DOI is permanent and version-specific.
- Software that isn't deposited anywhere depends on one person and one platform
- FAIR4RS (Findable, Accessible, Interoperable, Reusable) is the framework
- Public is not the same as shared: license, citation, and description do the work

Up next

Karla Padilla — Adding an Open-Source License

Laura Langdon — Creating a CITATION.cff File

Reid Otsuji — Minting a DOI and Creating a Release

Then back to Tim for Improving Metadata and Discoverability.